

- with LiDAR traces of a basement fault zone with poorly coalesced shear fracture clusters
- Friction tests on fault rocks and slip tendency analysis of mapped slip planes predict fault instability at observed hypocentral depths
 - This and other structurally studied seismic sources in the region manifest the hazards of prestressed, immature, subtle intraplate faults

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Intraplate regions commonly host energetic earthquakes on less-prominent faults, raising questions on how fault structure may influence intraplate seismogenesis. We investigate the causative fault of the 2024 Mw4.8 New Jersey earthquake which occurred near the misoriented 300-km long Ramapo Fault, by integrating seismicity-guided field geological mapping with friction experiments and slip tendency analysis. We find that the up-dip projection of relocated aftershocks aligns with a zone of NNE-striking topographic (LiDAR) traces of fault scarps, marking a previously unmapped fault zone in the New Jersey Highlands, henceforth designated as “Mountainville Fault (MF).” In basement outcrops along the topographic lineaments, we observe a predominance of NNE-striking/E-dipping, slickensided, gougeless fracture clusters with horsetail tip-splays, indicating a brittle shear zone origin. The fractures are poorly coalesced, and comprise steeply-dipping sets that are crosscut by gently-dipping sets. At the surface, MF lacks a break-through principal slip surface (core), and the geomorphic scarps and fractures strike at $\sim 15\text{--}25^\circ$ obliquity to the mainshock's nodal plane, suggesting that the field-mapped structures represent the Riedel-shears of a deeper blind fault core, characteristic of an “immature” fault zone. Laboratory-constrained frictional behavior of sampled representative fault rocks, and slip tendency computation from focal mechanism-resolved stress field predict frictional instability of the MF. Additionally, the friction experiment results suggest a possible contribution of concealed fluid-rock alterations at the depth of the mainshock. Altogether, these MF characteristics fit those of other structurally investigated earthquake sources in the region, demonstrating the seismic hazard of elusive, critically-oriented immature intraplate faults.

Plain Language Summary Studies of intraplate earthquake hazards commonly focus on mature, prominent faults. Yet, earthquake records document moderate-magnitude earthquakes on inconspicuous faults. The 2024 Mw4.8 Tewksbury, New Jersey earthquake occurred near the ~ 300 -km long Ramapo Fault, which, although constituting the largest regional fault, is misoriented for slip in the current tectonic stress regime. The upward projection of aftershocks aligns with the topographic (LiDAR) trace of a previously unmapped fault zone 3.5 km west of the epicenter. We follow this lineament in the field, and all the examined bedrock exposures show clusters of subparallel, gougeless, paleo-slip surfaces with steeply-dipping sets crosscut by gently-dipping sets. Relative to the strike of the mainshock's nodal plane, the fractures and LiDAR fault scarps define Riedel